Overview of Light Schrödinger Bridge

Let Ω be the state space, e.g., $C([0,1],\mathbb{R}^d)$, and let $\mathcal{P}(\Omega)$ be the space of all probability measures on Ω. For any measure $S \in \mathcal{P}(\Omega)$ denote its marginal distribution at $t = 0, 1$ as π^S and denote as $T_{|x_0,x_1}$ its bridges on the values x_0, x_1 at times $t = 0$, $t = 1$ respectively.

Let $W^\epsilon \in \mathcal{P}(\Omega)$ be the Wiener Process in Ω with volatility $\epsilon > 0$, which starts at the prescribed marginal $p_0 \in \mathcal{P}(\mathbb{R}^d)$ at $t = 0$, meaning it satisfies the following SDE:

$$\mathrm{d}W_t^\epsilon = \sqrt{\epsilon}\mathrm{d}W_t, \qquad W_0^\epsilon \sim p_0$$

Let $S^* \in \mathcal{P}(\Omega)$ be the unique solution of the Schrödinger problem (defined by the prescribed marginals $p_0, p_1 \in \mathcal{P}(\mathbb{R}^d)$ and the reference measure W^ϵ). Then it can disintegrated as:

$$S^*(\cdot) = \int_{\mathbb{R}^d\times\mathbb{R}^d} W^\epsilon_{|x_0,x_1}(\cdot)\pi^{S^*}(\mathrm{d}x_0, \mathrm{d}x_1)$$

where π^{S^*} is the the solution of the EOT problem with quadratic cost, on the space of all admissible plans $\Pi(p_0, p_1)$.

This measure is described by the following SDE:

$$\mathrm{d}Z_t = g^*(t, Z_t)\mathrm{d}t + \mathrm{d}W_t^\epsilon, \qquad Z_0 \sim p_0$$

where $g^* : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ is the optimal drift of the Schrödinger bridge S^*.

Furthermore, the optimal plan can also be written using the Schrödinger potentials $\psi^*, \phi^* : \mathbb{R}^d \to \mathbb{R}_+$, as:

$$\pi^{S^*}(x_0, x_1) = \psi^*(x_0)\exp\left\{-\frac{\|x_0 - x_1\|^2}{2\epsilon}\right\}\phi^*(x_1)$$

and the optimal drift can be derived from ϕ^* as:

$$g^*(t, x) = \epsilon\nabla_x \log \int_{\mathbb{R}^d} N(z\ ;\ x, (1-t)\epsilon\mathbf{I})\phi^*(\mathrm{d}z)$$

The main idea of the Light Schrödinger Bridge is to recover a parametric approximation $\pi_\theta \approx \pi^{S^*}$ which will be used to construct an approximation $S_\theta \approx S^*$ of the SB process (Light Schrödinger Bridge).

To learn π_θ we need to minimize the KL divergence between π^{S^*} and π_θ:

$$\theta^* = \arg\min_{\theta\in\Theta} \mathrm{KL}\left(\pi^{S^*} \mid \pi_\theta\right)$$

Define the adjusted Schrödinger potentials as:

$$u^*(x_0) = \exp\left\{-\frac{\|x_0\|^2}{2\epsilon}\right\}\psi^*(x_0), \quad v^*(x_0) = \exp\left\{-\frac{\|x_1\|^2}{2\epsilon}\right\}\phi^*(x_1)$$

And using them, the optimal coupling can be written as:

$$\pi^{S^*}(x_0, x_1) = u^*(x_0)\exp\left\{\langle x_0, x_1\rangle/\epsilon\right\}v^*(x_1)$$

which means that $\pi^{S^*}(x_1 \mid x_0) \propto \exp\left\{\langle x_0, x_1\rangle/\epsilon\right\}v^*(x_1)$.

We define the parameterization of $v^*(x_1)$ as $v_\theta(x_1)$ and with that we can define the parameterized coupling $\pi_\theta(x_0, x_1)$ from v_θ:

$$\begin{aligned}\pi_\theta(x_0, x_1) &= p_0(x_0)\pi_\theta(x_1 \mid x_0) \\ &= p_0(x_0)\frac{\exp\{\langle x_0, x_1\rangle/\epsilon\}v_\theta(x_1)}{c_\theta(x_0)}\end{aligned}$$

where c_θ is the normalization $c_\theta(x_0) = \int_{\mathbb{R}^d}\exp\{\langle x_0, x_1\rangle/\epsilon\}v_\theta(\mathrm{d}x_1)$.

In the original paper Light Schrödinger Bridge, they proved that for that parameterization it holds that the KL objective admits the representation:

$$\mathrm{KL}\left(\pi^{S^*} \mid \pi_\theta\right) = L(\theta) - L^*$$

where $L^* \in \mathbb{R}$ is a constant depending on p_0, p_1, and ϵ, and $L(\theta)$ is given by:

$$L(\theta) = \int_{\mathbb{R}^d}\log c_\theta(x_0)p_0(\mathrm{d}x_0) - \int_{\mathbb{R}^d}\log v_\theta(x_1)p_1(\mathrm{d}x_1)$$

The author notices that for $x_0 = 0$ $\pi^{S^*} \propto v^*(x_1)$, which means that v^* can be viewed as an unnormalized density. Thus they proposed to define it as a Gaussian Mixture, deriving the parameterization of the Schrödinger potential v_θ:

$$v_\theta(x_1) = \sum_{k=1}^{K} a_k N(x_1\ ;\ r_k, \epsilon S_K)$$

where the parameters are $\theta = \{a_k, r_k, S_k\}_{k=1}^K$, with $a_k \geq 0$, $r_k \in \mathbb{R}^d$, and $S_k \in \mathbb{R}^{d\times d}$ positive and symmetric.

Using that parameterization we can derive the conditional coupling:

$$\pi_\theta(x_1 \mid x_0) = \frac{1}{c_\theta(x_0)}\sum_{k=1}^{K}\bar{a_k}(x_0)N(x_1\ ;\ r_k + S_k x_0, \epsilon S_k)$$

where $\bar{a_k}(x_0) = a_k\exp\{(x_0^\top S_k x_0 + 2r_k^\top x_0)/(2\epsilon)\}$, and $c_\theta(x_0) = \sum_{k=1}^K \bar{a_k}(x_0)$.

Furthermore, let v_θ be an unnormalized Gaussian Mixture and π_θ be given by $p_\theta(x_0, x_1) = p_0(x_0)\pi_\theta(x_1 \mid x_0)$, then the measure S_θ with $S_{\theta|x_0,x_1} = W^\epsilon_{|x_0,x_1}$, and $\pi^{S_\theta} = \pi_\theta$, is a diffusion process governed by the following SDE:

$$\mathrm{d}X_t = g_\theta(t, X_t)\mathrm{d}t + \sqrt{\epsilon}\mathrm{d}W_t, \quad X_0 \sim p_0$$

where the drift g_θ can be described with the following closed form:

$$g_\theta(t,x) = \epsilon\nabla_x \log\left\{ N(x\ ;\ 0, \epsilon(1-t)\mathbf{I})\ \sum_{k=1}^{K}[a_k N(r_k\ ;\ 0, \epsilon S_k) N(h_k(t,x)\ ;\ 0, A_k^t)] \right\}$$

where the quantities $h_k(t,x)$, and A_k^t are given by:

$$h_k(t,x) = \frac{1}{\epsilon(1-t)}x + \frac{1}{\epsilon}S_k^{-1}r_k$$

$$A_k^t = \frac{t}{\epsilon(1-t)}\mathbf{I} + \frac{1}{\epsilon}S_k^{-1}$$

The last results the authors of Light Schrödinger Bridge proved is the universal approximation property of the Gaussian Mixture parameterization:

Assume that the prescribed marginals p_0, p_1 are compactly supported. Then for all $\delta > 0$ there exists a Gaussian mixture v_θ with scalar covariances $S_k = \lambda_k \mathbf{I} > 0$, that satisfies the inequality:

$$\mathrm{KL}(S^* \mid S_\theta) = \mathrm{KL}(\pi^{S^*} \mid \pi_\theta) < \delta$$

Overview of Optimal Schrödinger Bridge Matching

For every coupling $\pi \in \Pi(p_0, p_1)$ let T_π be the reciprocal process of π, i.e.,

$$T_\pi = \int_{\mathbb{R}^d \times \mathbb{R}^d} W^\epsilon_{|x_0,x_1}\pi(\mathrm{d}x_0, \mathrm{d}x_1)$$

Define, also, the reciprocal projection of an arbitrary measure $T \in \mathcal{P}(\Omega)$ as:

$$\mathrm{proj}_{\mathcal{R}}(T) = \int_{\mathbb{R}^d \times \mathbb{R}^d} W^\epsilon_{|x_0,x_1}\pi^T(\mathrm{d}x_0, \mathrm{d}x_1)$$

For any reciprocal process T_π, its Markovian projection $\mathrm{proj}_{\mathcal{M}}(T_\pi)$ is a diffusion process defined by the following SDE that preserves all time marginals of T_π (Light and Optimal Schrödinger Bridge Matching):

$$\mathrm{d}X_t = g(t, X_t)\mathrm{d}t + \sqrt{\epsilon}\mathrm{d}W_t$$

The drift is analytically given by:

$$g(t, x_t) = \int_{\mathbb{R}^d} \frac{x_1 - x_t}{1 - t} \, p_{T_\pi}(x_1 \mid x_t) \mathrm{d}x_1$$

which is the solution to the following minimization problem:

$$\min_g \left\{ \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{(x_0, x_1) \sim \pi, x_t \sim W^\epsilon_{|x_0, x_1}} \left[\left\| g(t, x_t) - \frac{x_1 - x_t}{1 - t} \right\|^2 \right] \right\}$$

Let $\mathcal{S}$ be the space of all Schrödinger Bridges:

$$\mathcal{S} = \{S \in \mathcal{P}(\Omega) : \text{there exist } p_0^S, p_1^S \in \mathcal{P}(\mathbb{R}^d), \text{ such that } S = \arg \min_{T \in \mathcal{F}(p_0^S, p_1^S)} \mathrm{KL}(T \mid W^\epsilon)\}$$

We define the optimal projection for a reciprocal process T_π as:

$$\mathrm{proj}_{\mathcal{S}}(T_\pi) = \arg \min_{S \in \mathcal{S}} \mathrm{KL}(T_\pi \mid S)$$

The authors of Light and Optimal Schrödinger Bridge Matching proved that for a given map $\pi \in \Pi(p_0, p_1)$ the optimal projection of a reciprocal process T_π leads to the Schrödinger Bridge S^* of p_0, p_1:

$$\mathrm{proj}_{\mathcal{S}}(T_\pi) = S^*$$

Let $\mathcal{S}(p_0) \subset \mathcal{S}$ be the subset of all Schrödinger Bridges that start at the marginal $p_0 \in \mathcal{P}(\mathbb{R}^d)$ at time $t = 0$. Since the Schrödinger Bridge we aim to approximate starts at p_0 at time $t = 0$, i.e., $S^* \in \mathcal{S}(p_0)$ then it suffices to optimize over $\mathcal{S}(p_0)$.

Let $S \in \mathcal{S}(p_0)$ be a Schrödinger Bridge starting at p_0 at time $t = 0$, and let $\pi \in \Pi(p_0, p_1)$ be admissible transport plan between p_0, and p_1. Then the optimal projection objective of its reciprocal process is derived as:

$$\mathrm{KL}(T \mid S) = C(\pi) + \frac{1}{2\epsilon} \mathbb{E}_{t \sim U[0,1],\, (x_0, x_1) \sim \pi,\, x_t \sim W^\epsilon_{|x_0, x_1}} \left[\left\| g_S(t, x_t) - \frac{x_1 - x_t}{1 - t} \right\|^2 \right]$$

where g_S is the Schrödinger drift of S, and $C(\pi)$ a quantity independent of S. Therefore the Schrödinger Bridge S^* is given as the solution to the following minimization problem:

$$S^* = \arg \min_{S \in \mathcal{S}(p_0)} \left\{ \frac{1}{2\epsilon} \mathbb{E}_{t \sim U[0,1],\, (x_0, x_1) \sim \pi,\, x_t \sim W^\epsilon_{|x_0, x_1}} \left[\left\| g_S(t, x_t) - \frac{x_1 - x_t}{1 - t} \right\|^2 \right] \right\}$$

Because every process $S \in \mathcal{S}(p_0)$ is uniquely determined by its terminal adjusted potential $v : \mathbb{R}^d \to \mathbb{R}_+$, and every valid potential v uniquely characterizes a velocity drift g_v, optimizing over the infinite-dimensional space of processes $\mathcal{S}(p_0)$ is strictly equivalent to optimizing over the space of potentials:

$$v^* = \arg\min_{v \in \mathcal{V}} \left\{ \frac{1}{2\epsilon} \mathbb{E}_{t\sim U[0,1],\ (x_0,x_1)\sim\pi,\ x_t \sim W^{\epsilon}_{|x_0,x_1}} \left[\left\| g_v(t,x_t) - \frac{x_1 - x_t}{1-t} \right\|^2 \right] \right\}$$

To make this optimization tractable, we restrict the search space to a parametric family of unnormalized Gaussian mixtures $v_\theta(x_1) = \sum_{k=1}^{K} \alpha_k N(x_1 \mid r_k, \epsilon S_k)$. This transforms the objective into a minimization over the finite-dimensional parameter space $\theta = \{\alpha_k, r_k, S_k\}_{k=1}^{K}$.

Under this specific Gaussian mixture parameterization, the drift g_θ resolves analytically into the exact closed-form expression:

$$g_\theta(t,x) = \epsilon \nabla_x \log \left\{ N(x\ ;\ 0, \epsilon(1-t)\mathbf{I}) \sum_{k=1}^{K} [a_k N(r_k\ ;\ 0, \epsilon S_k) N(h_k(t,x)\ ;\ 0, A_k^t)] \right\}$$

By substituting this exact parametric drift g_θ into the optimal projection framework, it can be shown that minimizing the mean squared error of the closed-form drift is mathematically identical to minimizing the KL divergence $\mathrm{KL}(\pi^{S^*} \parallel \pi_\theta)$.

Specifically, the regression objective evaluates identically to the dual Entropic Optimal Transport divergence up to an additive constant $\bar{C}(\pi)$ that is entirely independent of θ:

$$\frac{1}{2\epsilon} \mathbb{E}_{t\sim U[0,1],\ (x_0,x_1)\sim\pi,\ x_t \sim W^{\epsilon}_{|x_0,x_1}} \left[\left\| g_\theta(x_t,t) - \frac{x_1 - x_t}{1-t} \right\|^2 \right] = \bar{C}(\pi) + \mathrm{KL}\left(\pi^{S^*} \parallel \pi_\theta\right)$$

Proposition 1 (Universal Approximation of the Schrödinger Bridge)**.** *Assume the prescribed marginal distributions p_0 and p_1 are compactly supported. Let π^{S^*} be the true optimal plan of the Schrödinger Bridge, and let π_θ be the parameterized plan induced by an unnormalized Gaussian mixture potential with parameter set $\theta \in \Theta_K$ consisting of K components. Then, optimizing over the finite-dimensional parameter space Θ_K asymptotically recovers the global minimum over the infinite-dimensional space of all valid potentials, such that:*

$$\lim_{K\to\infty} \inf_{\theta\in\Theta_K} KL(\pi^{S^*} \mid \pi_\theta) = 0$$

Proof. Let $\mathcal{V}$ denote the infinite-dimensional space of all valid terminal adjusted potentials, and let $\mathcal{V}_\Theta = \bigcup_{K=1}^{\infty} \mathcal{V}_K$ denote the space of all unnormalized Gaussian mixture

potentials, where $\mathcal{V}_K$ represents the subset of potentials parameterized by exactly K components.
Because $\mathcal{V}_\Theta \subset \mathcal{V}$, to show that these optimizations are asymptotically equivalent, we rely on the universal approximation property of Gaussian mixtures.
Observe that any mixture of K components can be represented as a mixture of $K+1$ components by setting the weight of the additional component to zero $\mathcal{V}_K \subset \mathcal{V}_{K+1}$. Consequently, the sequence of infima over these spaces is monotonically non-increasing and bounded below by the non-negativity of the KL divergence:

$$\inf_{\theta \in \Theta_K} \mathrm{KL}(\pi^{S^*} \mid \pi_\theta) \geq \inf_{\theta \in \Theta_{K+1}} \mathrm{KL}(\pi^{S^*} \mid \pi_\theta) \geq 0$$

By the universal approximation property of Gaussian mixtures, for any arbitrary $\delta > 0$, there exists some finite number of components K_δ and a specific parameter set $\theta' \in \Theta_{K_\delta}$ such that:

$$\mathrm{KL}(\pi^{S^*} \mid \pi_{\theta'}) < \delta$$

Because the sequence of infima is non-increasing, it follows that for all $K \geq K_\delta$, the infimum over Θ_K is strictly bounded from above by δ:

$$0 \leq \inf_{\theta \in \Theta_K} \mathrm{KL}(\pi^{S^*} \mid \pi_\theta) \leq \mathrm{KL}(\pi^{S^*} \mid \pi_{\theta'}) < \delta$$

Since this sequence of infima is monotonically non-increasing, strictly non-negative, and can be bounded below any arbitrary $\delta > 0$ for a sufficiently large K, it mathematically satisfies the formal definition of a limit converging to zero:

$$\lim_{K \to \infty} \inf_{\theta \in \Theta_K} \mathrm{KL}(\pi^{S^*} \mid \pi_\theta) = 0$$

□

Algorithm 1: LightSB-M

Input: plan $\pi \in \Pi(p_0, p_1)$ accessible by samples, adjusted Schrödinger potential v_θ parameterized by a Gaussian mixture $\theta = \{a_k, r_k, S_k\}_{k=1}^K$.

while *Not Converged* **do**

Sample batch pairs $\{x_0^n, x_1^n\}_{n=1}^N \sim \pi$

Sample batch $\{t_n\}_{n=1}^N \sim U[0,1]$

Sample batch $\{x_t^n\}_{n=1}^N \sim W^\epsilon_{|x_0,x_1}$

$$L_\theta \leftarrow \frac{1}{N} \sum_{n=1}^N \left\| g_\theta(t_n, x_t^n) - \frac{x_1^n - x_t^n}{1 - t_n} \right\|^2$$

Update θ using $\partial_\theta L_\theta$

Output: Learned drift g_θ approximating the optimal g^*.

Algorithmic Implementation

The algorithm dictates that the loss is calculated as:

$$L(\theta) = \mathbb{E}_{t\sim U[0,1],\ (x_0,x_1)\sim\pi\ x_t\sim W^{\epsilon}_{|x_0,x_1}} \left[\left\| g_\theta(t,x_t) - \frac{x_1 - x_t}{1-t} \right\|^2 \right]$$

We can substitute the definition of the Brownian Bridge and write:

$$x_t = tx_1 + (1-t)x_0 + \sqrt{\epsilon t(1-t)}z, \quad z \sim N(0,\mathbf{I})$$

Therefore the original objective becomes:

$$L(\theta) = \mathbb{E}_{t\sim U[0,1],\ (x_0,x_1)\sim\pi\ x_t\sim W^{\epsilon}_{|x_0,x_1}} \left[\left\| g_\theta(t,x_t) - (x_1 - x_0) - \sqrt{\frac{\epsilon t}{1-t}}\, z \right\|^2 \right], \quad z \sim N(0,\mathbf{I})$$

As $t \to 1$, the variance of the target diverges to infinity due to the $\sqrt{\frac{\epsilon t}{1-t}}$ term, leading to severe training instability. To mitigate this without shifting the global minimum of the regression, we introduce a time-dependent weighting function $\lambda(t) = 1 - t$ into the objective:

$$L(\theta) = \mathbb{E}_{t\sim U[0,1],\ (x_0,x_1)\sim\pi\ x_t\sim W^{\epsilon}_{|x_0,x_1}} \left[\lambda(t) \left\| g_\theta(t,x_t) - (x_1 - x_0) - \sqrt{\frac{\epsilon t}{1-t}}\, z \right\|^2 \right]$$

By multiplying the squared error by $1 - t$, the effective variance of the noise target becomes bounded (ϵt), ensuring stable gradients throughout the entire integration interval $[0, 1]$.

Crucially, under the proposed Gaussian mixture parameterization for the terminal potential, the drift $g_\theta(t, x)$ is not a black-box neural network requiring backpropagation through complex layers. Instead, it is evaluated directly using its exact analytical form:

$$g_\theta(t,x) = \epsilon \nabla_x \log \left\{ N(x\ ;\ 0, \epsilon(1-t)\mathbf{I}) \sum_{k=1}^{K} [a_k N(r_k\ ;\ 0, \epsilon S_k) N(h_k(t,x)\ ;\ 0, A_k^t)] \right\}$$

where the quantities $h_k(t, x)$, and A_k^t are given by:

$$h_k(t,x) = \frac{1}{\epsilon(1-t)}x + \frac{1}{\epsilon}S_k^{-1}r_k$$

$$A_k^t = \frac{t}{\epsilon(1-t)}\mathbf{I} + \frac{1}{\epsilon}S_k^{-1}$$

Note that according to the paper, the second Gaussian form $N(h_k(t,x)\ ;\ 0, A_k^t)$ has positive exponent.

For a symmetric positive definite (SPD) covariance matrix S_k (like in our case) we can derive the following proposition.

Proposition 2 (Analytical Forward-Pass Drift)**.** *Let S_θ be the approximate Schrödinger Bridge process parameterized by an unnormalized Gaussian mixture terminal potential $v_\theta(x_1) = \sum_{k=1}^K a_k N(x_1 \mid r_k, \epsilon S_k)$. For any time $t \in [0,1)$ and state $x \in \mathbb{R}^d$, the closed-form drift $g_\theta(t,x)$ can be exactly evaluated as:*

$$g_\theta(t,x) = \frac{-x + \sum_{k=1}^K \Omega_k(t,x)(A_k^t)^{-1} h_k(t,x)}{1-t}$$

where the quantities $h_k(t,x)$ and A_k^t are defined as:

$$h_k(t,x) = \frac{1}{\epsilon(1-t)}x + \frac{1}{\epsilon}S_k^{-1} r_k$$
$$A_k^t = \frac{t}{\epsilon(1-t)}\mathbf{I} + \frac{1}{\epsilon}S_k^{-1}$$

and $\Omega_k(t,x)$ represents the normalized Softmax probability of the k-th mixture component evaluated at x:

$$\Omega_k(t,x) = \frac{w_k(t,x)}{\sum_{j=1}^K w_i(t,x)}$$

with $w_k(t,x)$ be given by:

$$w_k(t,x) = a_k N(r_k\ ;\ 0, \epsilon S_k)\bar{N}(h_k(t,x)\ ;\ 0, A_k^t)$$

where we write $\bar{N}(x\ ;\ \mu, S)$ the Gaussian at the variable x, with mean μ, and covariance matrix S, but with positive exponent.

Proof. We start from the general definition of the parameteric drift for a Gaussian mixture:

$$g_\theta(t,x) = \epsilon\nabla_x \log\left\{ N(x\ ;\ 0, \epsilon(1-t)\mathbf{I})\ \sum_{k=1}^K \left[a_k N(r_k\ ;\ 0, \epsilon S_k)\ \bar{N}(h_k(t,x)\ ;\ 0, A_k^t)\right]\right\}$$

We define the quantity inside the summation as w_k:

$$w_k(t,x) = a_k N(r_k\ ;\ 0, \epsilon S_k)\ \bar{N}(h_k(t,x)\ ;\ 0, A_k^t)$$

So the drift gets the form:

$$g_\theta(t,x) = \epsilon\nabla_x \log N(x\ ;\ 0, \epsilon(1-t)\mathbf{I}) + \epsilon\nabla_x \log \sum_{k=1}^{K} w_k(t,x)$$

For the first term we derive:

$$\epsilon\nabla_x N(x\ ;\ 0, \epsilon(1-t)\mathbf{I}) = -\frac{x}{1-t}$$

And for the second term we derive:

$$\epsilon\nabla_x \sum_{k=1}^{K} w_k(t,x) = \epsilon \sum_{k=1}^{K} \Omega_k(t,x)\nabla_x \log w_k(t,x)$$

where Ω_k is defined as the softmax:

$$\Omega_k(t,x) = \frac{w_k(t,x)}{\sum_{i=1}^{K} w_i(t,x)}$$

Therefore the parametric drift gets the following form:

$$g_\theta(t,x) = -\frac{x}{1-t} + \epsilon \sum_{k=1}^{K} \Omega_k(t,x)\nabla_x \log w_k(t,x)$$

By evaluating the gradient on the log expression of w_k we derive:

$$\nabla_x \log w_k(t,x) = \nabla_x \left\{ \frac{1}{2}\, h_k(t,x)^\top (A_k^t)^{-1} h_k(t,x) \right\}$$

We have defined S_k as a positive diagonal. Therefore its inverse exists. Because it is also SDP $(S_k)^{-1}$ is also SDP, thus $(A_k^t)^{-1}$ is SDP. This allows to write:

$$\begin{aligned}\nabla_x \left\{ \frac{1}{2}\, h_k(t,x)^\top (A_k^t)^{-1} h_k(t,x) \right\} &= (\nabla_x h_k(t,x))^\top (A_k^t)^{-1} h_k(t,x) \\ &= \frac{1}{\epsilon(1-t)} (A_k^t)^{-1} h_k(t,x)\end{aligned}$$

Which allow us to express the gradient of the log of w_k in closed form:

$$\nabla_x \log w_k(t,x) = \frac{1}{\epsilon(1-t)} (A_k^t)^{-1} h_k(t,x)$$

Finally, substituting everything back to the drift we close the proposition:

$$g_\theta(t,x) = -\frac{x}{1-t} + \frac{1}{1-t} \sum_{k=1}^{K} \Omega(t,x)(A_k^t)^{-1} h_k(t,x)$$

□